Name: V.Mahalakshmi
Reg No: 192521181
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Route Finding Problem: Finding the Path Cost

Aim

To apply the route finding problem to the given graph and find the path (and its cost) from the start city Coimbatore to the goal city Chennai.

Given graph (directed edges with cost)

Coimbatore to Mettupalayam = 37
Coimbatore to Palladam = 38
Coimbatore to Pollachi = 37
Coimbatore to Erode = 100
Palladam to Kangayam = 32
Kangayam to Karur = 61
Karur to Trichy = 74
Pollachi to Palani = 60
Palani to Dindigul = 57
Dindigul to Trichy = 97
Erode to Salem = 66
Salem to Vellore = 200
Trichy to Chennai = 320
Vellore to Chennai = 140

Problem formulation

Initial state: Coimbatore
Goal state: Chennai
Actions: travel from a city to the next connected city along an arrow
Path cost: sum of the edge costs along the route

Procedure

1. Start at Coimbatore and list every route that leads to Chennai.
2. Mettupalayam has no outgoing road, so it is a dead end and gives no route to Chennai.
3. Add up the edge costs along each route to get its path cost.
4. Choose the route with the least path cost.

Possible routes and their path cost

Route 1: Coimbatore - Palladam - Kangayam - Karur - Trichy - Chennai

Cost = 38 + 32 + 61 + 74 + 320 = 525

Route 2: Coimbatore - Pollachi - Palani - Dindigul - Trichy - Chennai

Cost = 37 + 60 + 57 + 97 + 320 = 571

Route 3: Coimbatore - Erode - Salem - Vellore - Chennai

Cost = 100 + 66 + 200 + 140 = 506

Comparison

Route 1 = 525, Route 2 = 571, Route 3 = 506. The smallest value is 506, so Route 3 is the cheapest route.

Result

Best path: Coimbatore - Erode - Salem - Vellore - Chennai
Minimum path cost: 506

Note

The label 61 appears twice near the Kangayam to Karur edge in the figure. It has been taken as a single edge of cost 61.